Chapter 4

Expectation

4.1 The Expectation of a Random Variable

Commentary

It is useful to stress the fact that the expectation of a random variable depends only on the distribution of the random variable. Every two random variables with the same distribution will have the same mean. This also applies to variance (Sec. 4.3), other moments and m.g.f. (Sec. 4.4), and median (Sec. 4.5). For this reason, one often refers to means, variance, quantiles, etc. of a distribution rather than of a random variable. One need not even have a random variable in mind in order to calculate the mean of a distribution.

Solutions to Exercises

1. The mean of X is

$$E(X) = \int x f(x) dx = \int_a^b \frac{x}{b-a} dx = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

2. $E(X) = \frac{1}{100}(1 + 2 + \cdots + 100) = \frac{1}{100}\frac{(100)(101)}{2} = 50.5.$

3. The total number of students is 50. Therefore,

$$E(X) = 18\left(\frac{20}{50}\right) + 19\left(\frac{22}{50}\right) + 20\left(\frac{4}{50}\right) + 21\left(\frac{3}{50}\right) + 25\left(\frac{1}{50}\right) = 18.92.$$

4. There are eight words in the sentence and they are each equally probable. Therefore, the possible values of X and their probabilities are as follows:

x	$f(x)$
2	1/8
3	5/8
4	1/8
9	1/8

It follows that $E(X) = 2\left(\frac{1}{8}\right) + 3\left(\frac{5}{8}\right) + 4\left(\frac{1}{8}\right) + 9\left(\frac{1}{8}\right) = 3.75.$

5. There are 30 letters and they are each equally probable:

   2 letters appear in the only two-letter word;

   15 letters appear in three-letter words;

   4 letters appear in the only four-letter word;

   9 letters appear in the only nine-letter word.

   Therefore, the possible values of Y and their probabilities are as follows:

y	$g(y)$
2	2/30
3	15/30
4	4/30
9	9/30

$$E(Y) = 2\left(\frac{2}{30}\right) + 3\left(\frac{15}{30}\right) + 4\left(\frac{4}{30}\right) + 9\left(\frac{9}{30}\right) = 4.867.$$

6. $E\left(\dfrac{1}{X}\right) = \displaystyle\int_0^1 \frac{1}{x} \cdot 2x dx = 2.$

7. $E\left(\dfrac{1}{X}\right) = \displaystyle\int_0^1 \frac{1}{x}\,\mathrm{dx} = -\lim_{\mathrm{x}\to 0} \log(\mathrm{x}) = \infty$. Since the integral is not finite, $E\left(\dfrac{1}{X}\right)$ does not exist.

8. $E(XY) = \displaystyle\int_0^1 \int_0^x xy \cdot 12y^2 dy dx = \frac{1}{2}.$

9. If X denotes the point at which the stick is broken, then X has the uniform distribution on the interval $[0, 1]$. If Y denotes the length of the longer piece, then $Y = \max\{X, 1 - X\}$. Therefore,

$$E(Y) = \int_0^1 \max(x, 1-x)dx = \int_0^{1/2} (1-x)dx + \int_{1/2}^1 x dx = \frac{3}{4}.$$

10. Since α has the uniform distribution on the interval $[-\pi/2, \pi/2]$, the p.d.f. of α is

$$f(\alpha) = \begin{cases} \dfrac{1}{\pi} & \text{for } -\dfrac{\pi}{2} < \alpha < \dfrac{\pi}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

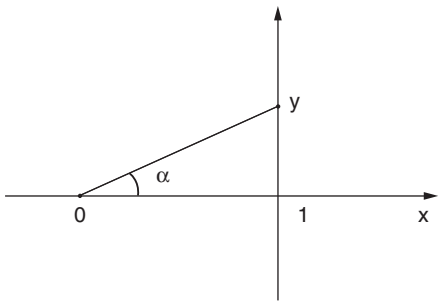


Figure S.4.1: Figure for Exercise 10 of Sec. 4.1.

Also, $Y = \tan(\alpha)$. Therefore, the inverse transformation is $\alpha = \tan^{-1} Y$ and $d\alpha/dy = 1/(1+y^2)$. As α varies over the interval $(-\pi/2, \pi/2)$, Y varies over the entire real line. Therefore, for $-\infty < y < \infty$, the p.d.f. of Y is

$$g(y) = f(\tan^{-1} y)\frac{1}{1+y^2} = \frac{1}{\pi(1+y^2)}.$$

11. The p.d.f.'s of Y_1 and Y_n were found in Sec. 3.9. For the given uniform distribution, the p.d.f. of Y_1 is

$$g_1(y) = \begin{cases} n(1-y)^{n-1} & \text{for} \quad 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$E(Y_1) = \int_0^1 yn(1-y)^{n-1} dy = \frac{1}{n+1}.$$

The p.d.f. of Y_n is

$$g_n(y) = \begin{cases} ny^{n-1} & \text{for} \quad 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$E(Y_n) = \int_0^1 y \cdot ny^{n-1} dy = \frac{n}{n+1}.$$

12. It follows from the probability integral transformation that the joint distribution of $F(X_1), \ldots, F(X_n)$ is the same as the joint distribution of a random sample from the uniform distribution on the interval $[0, 1]$. Since $F(Y_1)$ is the smallest of these values and $F(Y_n)$ is the largest, the distributions of these variables will be the same as the distributions of the smallest and the largest values in a random sample from the uniform distribution on the interval $[0, 1]$. Therefore, $E[F(Y_1)]$ and $E[F(Y_n)]$ will be equal to the values found in Exercise 11.

13. Let $p = \Pr(X = 300)$. Then $E(X) = 300p + 100(1-p) = 200p + 100$. For risk-neutrality, we need $E(X) = 110 * (1.058) = 116.38$. Setting $200p + 100 = 116.38$ yields $p = 0.0819$. The option has a value of 150 if $X = 300$ and it has a value of 0 if $X = 100$, so the mean of the option value is $150p = 12.285$. The present value of this amount is $12.285/1.058 = 11.61$, the risk-neutral price of the option.

14. For convenience, we shall not use dollar signs in these calculations.

    (a) We need to check the investor's net worth at the end of the year in four situations:

        i. $X = 180$ and she makes the transactions
        ii. $X = 180$ and she doesn't make the transactions
        iii. $X = 260$ and she makes the transactions
        iv. $X = 260$ and she doesn't make the transactions

Since we don't know the investor's entire net worth, we shall only calculate it relative to all other investments. This means that we only need to pretend as if the investor had one share of the stock worth 200. We don't care what else she has. We need to show that cases (i) and (ii) lead to the same net worth and that cases (iii) and (iv) lead to the same net worth. In case (ii), her net worth will change by -20. In case (iv), her net worth will change by 60. In case (i), nobody will exercise the options. So she will sell the three extra shares for 180 each (total 540) and pay the loan of 519.24 plus interest 20.77 for a net 0.01 loss. Plus her one original share of stock has lost 20 and her net worth has changed a total of -20.01, which is the same as case (i) except for the accumulated rounding error. In case (iii), the options will be exercised, and she will receive 800 for four shares of the stock. She will have to pay back the loan of 519.24 plus 20.77 in interest for a net gain of 259.99. But she no longer has the one share of stock that was worth 200, so her change in net worth is 59.99, the same as case (iv) to within the same one cent of rounding.

(b) If the option price is $x < 20.19$, then the investor only receives $4x$ for selling the options, but still needs to pay 600 for the three shares, so she must borrow $600 - 4x$. The rest of the calculations proceed just as in part (a) but we must replace 519.24 by $600 - 4x$, and the interest 20.77 must be replaced by $0.04(600 - 4x)$. That is, to pay back the loan with interest, she must pay $624 - 4.16x$ instead of 540.01. So she pays an additional $83.99 - 4.16x$ relative to the situation in case (a) regardless of what happens to the stock price.

(c) This situation is the same as part (b) except now the value of $83.99 - 4.16x$ is negative instead of positive, so the investor pays back less and hence makes additional profit rather than suffers additional loss.

15. The value of the option is 0 if $X = 260$ and it is 40 if $X = 180$, so the expected value of the option is $40(1 - p) = 40 \times 0.65 = 26$. The present value of this amount is $26/1.04 = 25$.

16. If f is the p.f. of X, and $Y = |X|$, then for $y \geq 0$, $\Pr(Y = y) = \Pr(X = y) + \Pr(X = -y)$. In Example 4.1.4, $\Pr(X = y) = \Pr(X = -y) = 1/[2y(y + 1)]$, and this makes $\Pr(Y = y)$ the p.f. in Example 4.1.5.

4.2 Properties of Expectations

Commentary

Be sure to stress the fact that Theorem 4.2.6 on the expected value of a product of random variables has the condition that the random variables are independent. This section ends with a derivation for the expectation of a nonnegative discrete random variable. Although this method has theoretical interest, it is not central to the rest of the text.

Solutions to Exercises

1. The random variable Y is equal to $10(R - 1.5)$ in dollars. The mean of Y is $10[E(R) - 1.5]$. From Exercise 1 in Sec. 4.1, we know that $E(R) = (-3 + 7)/2 = 2$, so $E(Y) = 5$.

2. $E(2X_1 - 3X_2 + X_3 - 4) = 2E(X_1) - 3E(X_2) + E(X_3) - 4 = 2(5) - 3(5) + 5 - 4 = -4$.

3.

$$\begin{aligned} E[(X_1 - 2X_2 + X_3)^2] &= E(X_1^2 + 4X_2^2 + X_3^2 - 4X_1X_2 + 2X_1X_3 - 4X_2X_3) \\ &= E(X_1^2) + 4E(X_2^2) + E(X_3^2) - 4E(X_1X_2) \\ &\quad + 2E(X_1X_3) - 4E(X_2X_3). \end{aligned}$$